Magnetic anisotropy and de Haas – van Alphen oscillations in a Bi micro- and nanowire arrays studied via cantilever magnetometry

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Measurements: Oscillatory magnetization at T = 0.5 K (magnetic fields up to 9 T) using silicon cantilever magnetometer

Samples: High density array of 50μm diameter Bi polycrystalline wires (5 mg); high density array of 200 nm diameter Bi oriented nanowires, wire axis along trigonal axis (1 mg).

Motivation: Long term goal is to apply sensitive magnetometry to study modifications of Fermi surface by nanometer scale confinement.

(1) Introduction

a) Extremely small effective masses, long Fermi wavelength make Bi ideal for studying quantum confinement effects

b) Proposed semi-metal to semiconductor transition (SMSC) for confinement < 60 nm

c) We have fabricated hiqh quality, densely packed arrays of nanowires[1]

d) Earlier measurements via oscillatory resistance (Shubnikov-de Hass effect) indicate that Fermi surface is strongly modified in nanowires, masking SMSC. Interpretation complicated by electrical contact difficulties[2,3]

e) GOAL: use cantilever magnetometer to probe confinement effects on Fermi surface via oscillatory magnetization (deHass-van Alphen effect, dHvA)[4]

(2) Cantilever magnetometer

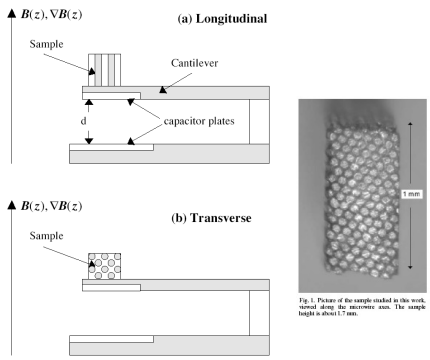
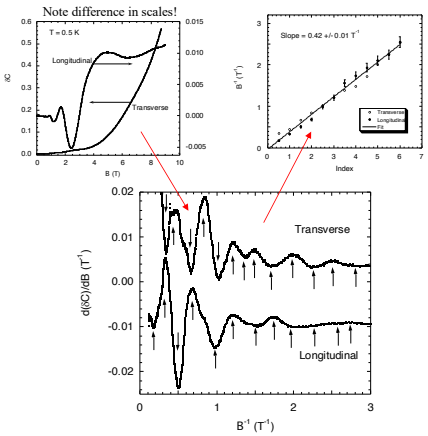


Fig. 1. Picture of the sample studied in this work, viewed along the microwire axis. The sample height is about 1.7 mm.

Small displacements:

δC = ΔC(H)/C(0) α M(H)H

(3) dHvA in 50 μm polycrystalline wires



(4) Discussion of 50 μm results

a) Transverse signal much larger, due to torque contribution

b) Clearly distinguish dHvA oscillation

c) Due to polycrystalline structure, analysis of oscillations is difficult

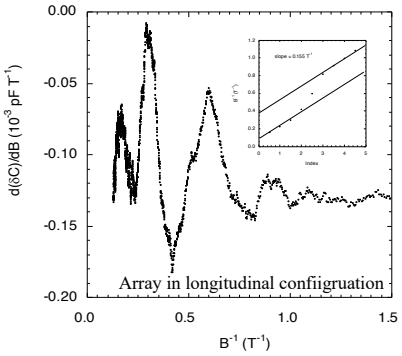
d) Primary period consistent with light electrons, indicating that larger orbits dominate the susceptibility

e) Dingle temperature is larger than 4 K, much larger than for bulk Bi or Bi thin films

f) Anticipated difficulties with nanowire arrays:
- will lose contribution of light electrons to dHvA for diameters less than cyctron radius
- smaller mass of samples

(5) Preliminary: 200 nm oriented wires

- smaller amplitude, collision damping?
- unusual shape of index plot; BUT, slopes consistent with SdH on 200nm wires



(6) Conclusions / References

(1) Bi microwire arrays exhibit strongly anistropic response due to geometric structure
(2) de Hass- van Alphen oscillations dominated by light holes with large cyclotron orbits
(3) Observation of dHvA oscillations in 200nm wire array
(a) Oscillations damped -- collisions with walls?
(b) Period consistent with SdH

(7) References

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[3] T. E. Huber et al., Appl. Phys. Lett. **84**, 1326 (2004)
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Funding: National Science Foundation Grant DMR0072847 and Army Research Office grant DAAD4006-MS-SAH